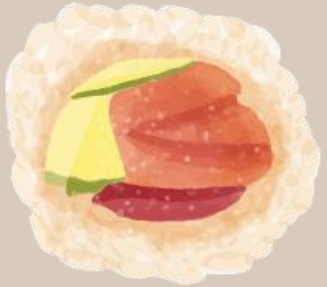


SUSHI ASSEMBLY LINE

Midterm Updates







MEET THE TEAM



NICOLAS JAO

SMART Systems Automation

Previous Experience:

Electrical Design Co-op at Mecsmart Systems



JACOB KWAN

SMART Systems Automation

Previous Experience:

Automation Service Solutions Co-op at Eclipse Automation



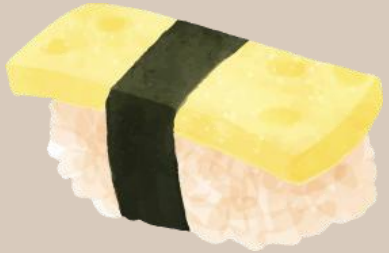
JAMIE WILLIAMS

Industrial Automation

Previous Experience:

Engineering Intern at Auper Electric Motorcycles

01



OUR PROBLEM STATEMENT

THE PROBLEM

SCENARIO

Basic sushi rolls are high demand at sushi restaurants, but takes long to make

VALUE

Automating menial, repetitive tasks frees the skilled chefs' time, allowing them to focus on more complex dishes

TARGET

For use in sushi restaurants or mass-produced sushi assembly lines for grocery stores





02

OUR OBJECTIVES AND DESIGNS

OBJECTIVES

HIGH-QUALITY

Produce consistent rolls that are uniform in shape

01





02

FOOD-GRADE

Equipment used allows for safe consumption

SAFE

Humans can safely interact with the parts of the assembly line

03





04

EFFICIENT

Rolls are produced continuously and quickly

5 STATIONS:

00 *Foundation & Set Up*

01 *Laying the Sheet of Rice*

02 *Dispensing the Neta*

03 *Rolling the Sushi*

04 *Cutting the Roll*

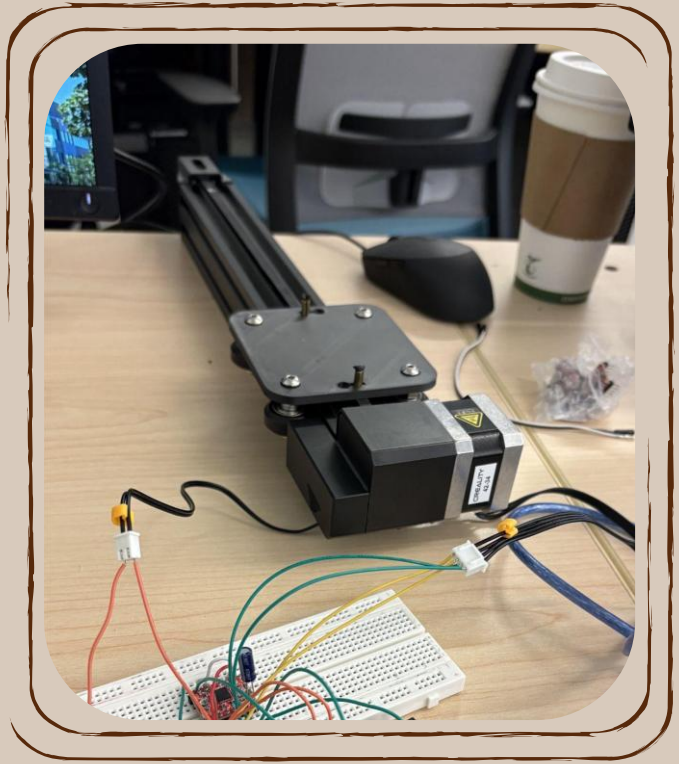
05 *Plating the Pieces*



0. LINEAR MOVING PLATFORM

Function: To carry the sushi roll from station to station across the table.

- Belt driven
- Uses a 42-34 Stepper motor
- Controlled with an A4988 stepper motor driver
- Uses 40x40mm V-slot aluminum extrusions
- ~1 meter of total movement
- Uses limit switches to detect platform position



1. DISPENSING SHEET OF SUSHI RICE



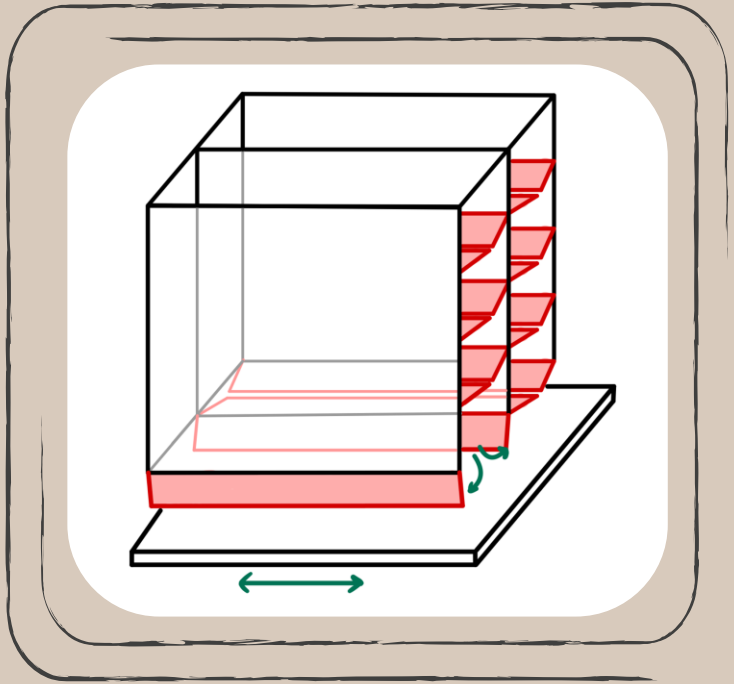
Function: Contains the rice stock and forces the rice through rollers to press into a sheet.

- Utilizes 2 silicone ‘rolling pins’
- Has two ‘agitators’ to push rice through the rollers
- Platform moves underneath, in tandem with the dispensed rice

2. LAYING OUT INGREDIENTS

Function: Drops the selected ingredients onto the rice sheet in perfect lines.

- An array of 8 (4x2) compartments for different ingredients
- Compartments: ~13cm x ~4cm
- Will only ever drop the lowest compartment or either column
- Allows for different ingredient combinations



3. ROLLING



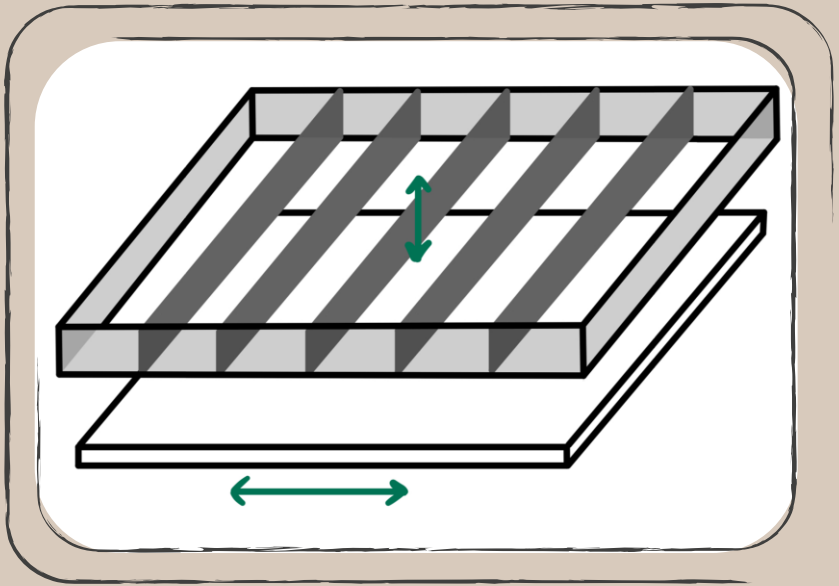
Function: Collapses inwards to roll the sushi roll on a bamboo mat.

- Uses a “ribcage” design with prongs that collapse into the roll from the bottom
- A high-torque stepper motor will control one set of ribs, and the other will follow the opposite direction through a gear system
- This platform will be mounted on the linear slide platform and held up by long spacers to give room to the ribs in rest position

4. CUTTING THE ROLL

Function: Slice the sushi roll into 6 evenly sliced pieces.

- Uses 5 evenly spaced, thin blades parallel to one another held within a frame
- Will have a plastic safety guard
- Moves straight up and down within metal frame
- Driven by stepper motor



5. PLATING THE ROLLS

Function: A robot picks and places the finished sushi pieces onto a plate, ready to serve.

- A UR robot with a gripper end-effector will pick and place the finished pieces on the platform one by one
- It will plate them onto a plate in a preprogrammed sequence
- Interfacing the robot controller with the main system controller is on our to-do list
- Learning how to program the robot is on our to-do list









SIMILAR PRODUCTS

03



AUTEC NORIMAKI MACHINE

- Good reference for our station (1) the rice rolling machine
- Designed to be used along with manually assembled rolls
- $21,000 CAD



SUSHI FACTORY ROLLING MECHANISMS

This mechanism uses a flexible material to wrap the roll together into a tube.



This design is much closer to our design but uses flat sides instead of curved ribs.





04

WHAT WE'VE ACHIEVED

THE TIMELINE

WEEKS 1-3

- *Brainstorming project ideas*
- *Assessing and narrowing ideas*
- *Finalizing project idea*
- *Conceptualizing initial concepts*

WEEK 4

- *Presenting proposal*

WEEKS 5-6

- *Narrowing down design concepts*
- *Developing design prototypes*
- *Start concept testing*

WEEK 7

- *Midterm report and presentation*

WEEKS 8-10

- *Form bill of materials*
- *Order parts*
- *Building prototypes of design components*

WEEK 11

- *Create poster for presentation*

WEEK 12

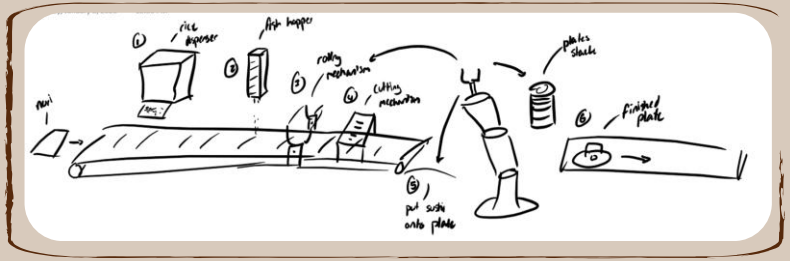
- *Finalize work for the term*
- *Write final report*

WEEK 13

- *Poster presentations*
- *Submitting final report*



PROGRESS ON OPERATION DESIGN



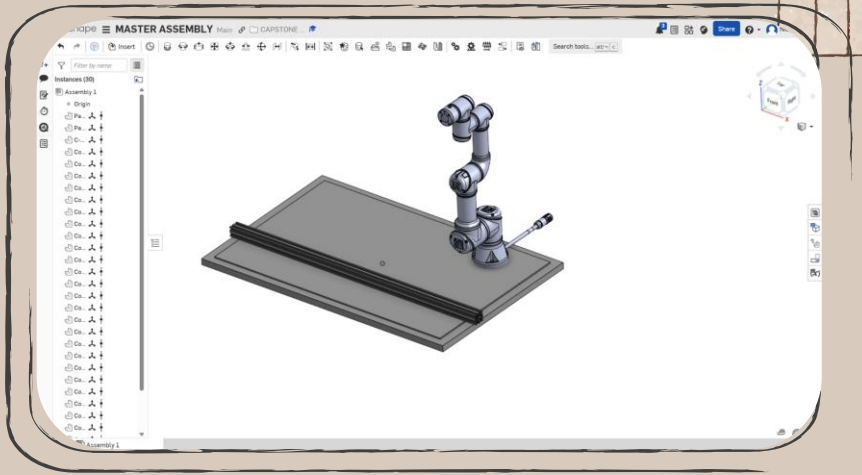
We went through each substation one by one to finalize a principle of operation with individual specifications.

- In the beginning, we had rough sketches for our ideas
- Now we are making progress on turning these rough sketches into real specifications (dimensions and materials)
- Testing these individual designs to see if they are effective is on our to-do list

PROGRESS ON CAD DESIGN

We made a master assembly file with the table as our base and began planning dimensions and spacing.

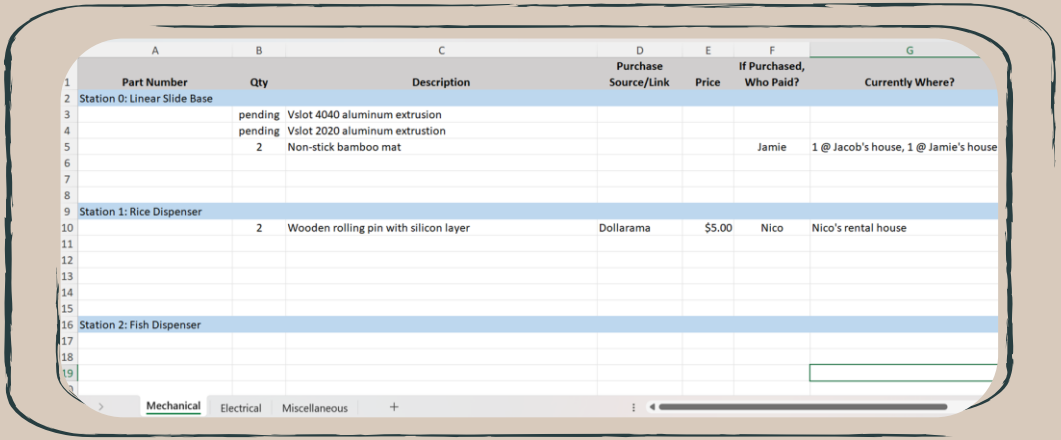
- We have a folder for imported parts and a folder for custom-designed parts
- We plan to have assemblies for each substation, then mate them all together in this master assembly



PROGRESS ON BILL OF MATERIALS

We made a master Excel file BOM.

- We split the BOMs into mechanical, electrical, and miscellaneous BOMs
- We started shopping for parts and CAD files online for these parts
- This file will keep track of who paid for what so we can be reimbursed properly at the end
- It also contains locations for all materials since we may have some at home or at ETB



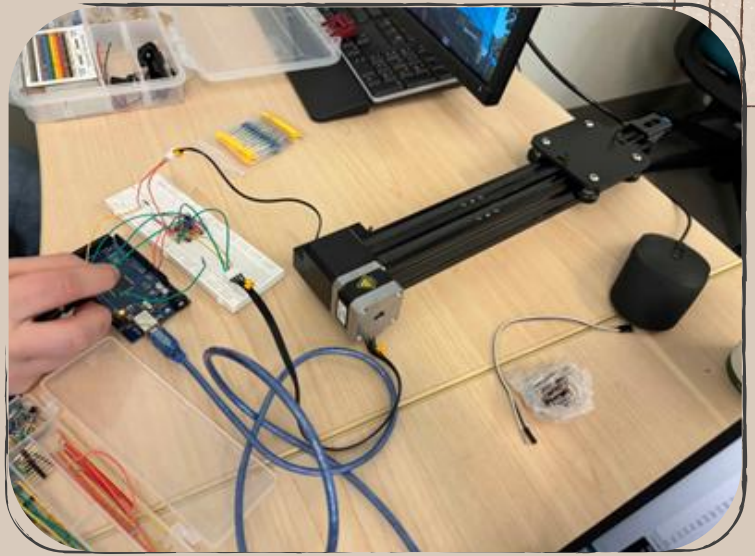
	A	B	C	D	E	F	G
1	Part Number	Qty	Description	Purchase Source/Link	Price	If Purchased, Who Paid?	Currently Where?
2	Station 0: Linear Slide Base						
3		pending	Vslot 4040 aluminum extrusion				
4		pending	Vslot 2020 aluminum extrustion				
5		2	Non-stick bamboo mat			Jamie	1 @ Jacob's house, 1 @ Jamie's house
6							
7							
8							
9	Station 1: Rice Dispenser						
10		2	Wooden rolling pin with silicon layer	Dollarama	$5.00	Nico	Nico's rental house
11							
12							
13							
14							
15							
16	Station 2: Fish Dispenser						
17							
18							
19							

Mechanical | Electrical | Miscellaneous

PROGRESS ON PROTOTYPING & TESTING

We tested the control of our stepper motors with a motor driver and an Arduino.

- We used an Arduino Uno and an L298N motor driver
- We corrected issues with voltage supply until we were satisfied with the control
- A proof-of-concept for the intended implementation of our motors



REFERENCES

- https://www.maisonrondeau.com/en/product/autec-asm865a-norimaki-machine-maki-sushi